ESE 4007 One-Shot Notes

Smart Grid Technology · KUET · past papers 2020, 2022, 2023 · answers from standard textbooks (no course slides available)

One-shot notes: ESE 4007 Smart Grid Technology

Smart grid in one line

Old grid + sensors (eyes) + two-way communication (nerves) + automatic control (brain). Digital, two-way power and data, distributed generation, self-healing, active customers.

Drivers: demand, ageing assets, reliability, renewables, climate, efficiency, customer choice, ICT. Bangladesh: losses/theft, peak gap, blackouts, renewables target, costly imported fuel.

Stakeholders

Generators, transmission/system operator (PGCB, NLDC), distribution utilities (DPDC, DESCO, BREB), consumers/prosumers, regulator (BERC, SREDA), vendors/ICT, aggregators, researchers/standards bodies.

Active vs passive distribution

Passive: no DG, one-way, fit and forget, simple protection. Active: DG/storage/flexible loads, two-way, real-time control.

Loop: measure → communicate → state estimate + forecast → control (Volt/VAR, DG dispatch, reconfiguration, DR, FLISR) → act.

Three technology families

ICT: fibre, PLC, ZigBee, Wi-Fi, WiMAX, cellular; protocols IEC 61850, DNP3. Sensing/control: smart meters, PMU, sensors, IEDs, SCADA/DMS, automation. Power electronics: inverters, FACTS, HVDC; storage: batteries, pumped hydro, flywheel, supercapacitor, SMES, CAES.

DG and microgrid

DG = small (kW-50 MW) generation in the distribution network near loads. Advantages: lower losses, better voltage, reliability, deferred investment, renewables, CHP, quick.

Microgrid: DGs + storage + loads + MGCC + static switch at PCC. On-grid: grid sets V, f; DGs in PQ mode. Islanded: switch opens, V/f or droop control, shed non-critical loads, resynchronise later.

Communication

SCADA: field devices/RTU/IED → network → master station + HMI. Advantages: real-time monitoring, remote control, faster restoration, efficiency, data, safety.

HAN (home, 10-100 m: ZigBee, Bluetooth, Wi-Fi) → NAN (neighbourhood, 1-10 km: RF mesh, PLC, WiMAX) → WAN (backbone: fibre, WiMAX, LTE).

Smart meter and AMI

Meter → NAN → concentrator → WAN → head-end → MDMS → billing, OMS, DMS, portal.

AMR one-way billing reads; AMI two-way: interval data, remote connect/disconnect, ToU and prepaid, outage alerts, HAN link.

DSM

Change timing/amount of use to flatten the load curve. Peak clipping (cut peak: direct load control), valley filling (add off-peak load: night EV charging), load shifting (move peak to off-peak: ToU, thermal storage). Plus conservation, load growth, flexible load.

PMU and WAMS

Measures V, I phasors + frequency + ROCOF, GPS time-stamped (1 µs), 25-60 per second. Blocks: CT/PT inputs, anti-aliasing filter, A/D, GPS + phase-locked oscillator, DFT microprocessor, modem.

WAMS: PMUs → local PDC → super PDC → monitoring, state estimation, protection.

Load flow

Buses: slack (V, δ), PV (P, $|V|$), PQ (P, Q). Methods: Gauss-Seidel (simple, slow), Newton-Raphson (fast convergence), fast decoupled (fastest), DC (linear).

Smart grid needs: stochastic, three-phase unbalanced, DG and two-way flow, high R/X robust, real-time with PMU/AMI data, large-scale.

Analysis tools

Contingency = what if one element trips (N-1). SSA = steady post-contingency limits check: secure / alert / emergency. SCOPF = cheapest dispatch still secure after contingencies. DSOPF = OPF + time + uncertainty (adaptive dynamic programming). Congestion = line at its limit: re-dispatch, FACTS, pricing. GIS: asset maps, outages, crews, planning.

Multi-agent systems

Agent: autonomous, reactive, pro-active, social. IDAPS agents: control (island/reconnect), DER, user (critical loads), database. Meter MAS: meter agents → collector agents → billing, database, monitoring agents.

Y_{bus}

Y_{ii} = sum of admittances at bus i (including shunt); $Y_{ij} = -y_{ij}$. KCL at each bus ⇒ $I_i = \Sigma Y_{ij}V_j$ ⇒ $I_{bus} = Y_{bus}V_{bus}$. Symmetric, sparse.

Infinite bus: constant V and f whatever the power.

Frequency control

Frequency = generation-load balance. Governor chain: fly-ball → linkage → hydraulic amplifier → valve; droop R leaves $\Delta f = -R\Delta P_L/(1 + DR)$.

LFC loop: governor $1/(1 + sT_g)$ → turbine $1/(1 + sT_t)$ → $1/(2Hs + D)$, feedback 1/R. AGC: integral of ACE = $\Delta P_{tie} + B\Delta f$; restores 50 Hz and tie flows; essential with renewables.

Reactive power and voltage

P ↔ frequency, Q ↔ voltage. Q devices: AVR/excitation, shunt capacitors/reactors, series capacitors, synchronous condensers, OLTC, SVC, STATCOM, smart inverters.

AVR: sensor → compare with V_{ref} → amplifier → exciter → field. SVC = TCR + TSC + filters. P_{max} = EV/X: raise E, V; cut X.

Rotor-angle stability

Steady-state = small slow changes; transient = large sudden faults. Swing: $(H/\pi f_0)d^2\delta/dt^2 = P_m - P_{max}\sin\delta$.

Small disturbance: $P_s = P_{max}\cos\delta_0$, $s = \pm j\sqrt{(\pi f_0 P_s/H)}$ ⇒ $\omega_n = \sqrt{(\pi f_0 P_s/H)}$; $P_s < 0$ unstable.

EAC: stable if A_2 (decelerating) can equal A_1 (accelerating) before $\delta_{max} = 180° - \delta_0$. $\cos\delta_{cr} = (\pi - 2\delta_0)\sin\delta_0 - \cos\delta_0$.

Machine numerical (2020)

$I = 0.625\angle -36.87°$. No saliency: $E' = 1 + j0.3I = 1.1226\angle 7.68°$, $P = 3.742\sin\delta$.

Saliency: $E_Q = 1 + j0.6I = 1.2612\angle 13.76°$; $I_d = 0.4832$; $E'_q = 1.1162$; $P = 3.721\sin\delta - 0.833\sin2\delta$.

Voltage stability

Causes: heavy load, long lines, Q shortage/generator limits, outages, OLTC action, motor loads. Prevent: Q reserves, capacitors/SVC, generator V set-points, block OLTC, re-dispatch, load shedding, PMU monitoring.

Techniques: P-V, Q-V, CPF, modal analysis, indices (L-index, VIPI), dynamic simulation, PMU/AI. VIPI = angle between load vector and the midpoint of the two load-flow solutions → 0 at collapse.

Power quality

Sag, swell, interruption, transients, harmonics, flicker, unbalance, frequency deviation. Matters for sensitive loads, heating/losses, inverter-rich grid. PQ audit: measure, find source, fix, comply.

Intelligence tools

ANN: layers of weighted neurons trained on data; forecasting, security assessment, fault detection, theft detection. ES: knowledge base + inference engine; alarm processing, fault diagnosis, restoration.

AHP: hierarchy → pairwise 1-9 → weights (eigenvector) → CR < 0.1 → synthesis. Computational challenges: size, uncertainty, real-time, non-linear optimisation, multi-time-scale, data/cyber, models.

Cyber security

Interoperability = exchange and use information (IEC 61850, CIM, DNP3, IEEE 2030). Threats: malware, DoS, man-in-the-middle, meter tampering, FDI, GPS spoofing; huge attack surface.

FDI: a = Hc passes the residual test → wrong state, wrong dispatch, overloads. Mitigation: defence in depth, authentication, encryption, protected key meters, PMUs, anomaly detection, response plans, training.

Renewables, V2G, barriers

Renewable challenges: variable, uncertain, low inertia, distributed (forecasting, storage, protection, PQ, markets). V2G: bidirectional charger, charge off-peak, discharge at peak, frequency regulation; battery wear.

Barriers: cost, regulation, standards, security, skills, consumer acceptance → phased rollout, policy, standards, training, awareness.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
A/D	analog-to-digital
ACE	area control error
ACL	agent communication language
ADB	Asian Development Bank
ADMS	advanced distribution management system
AGC	automatic generation control
AHP	analytical hierarchy process (the paper calls it analytical hierarchical programming)
AI	artificial intelligence
AMI	advanced metering infrastructure
AMR	automatic meter reading
ANN	artificial neural network
ANSI	American National Standards Institute
AVR	automatic voltage regulator
BAN	building area network
BERC	Bangladesh Energy Regulatory Commission
BREB	Bangladesh Rural Electrification Board
CAES	compressed air energy storage
CHP	combined heat and power
CI	consistency index
CIA	confidentiality, integrity, availability
CIM	common information model
CIP	critical infrastructure protection
CIS	customer information system
CO_2	carbon dioxide
COSEM	Companion Specification for Energy Metering
CPF	continuation power flow
CR	consistency ratio
CT	current transformer
DER	distributed energy resource
DESCO	Dhaka Electric Supply Company
DFT	discrete Fourier transform
DG	distributed generation
DLMS	Device Language Message Specification
DLR	dynamic line rating
DMS	distribution management system
DMZ	demilitarised zone (buffer network)

DNP3	Distributed Network Protocol 3
DoS	denial of service
DPDC	Dhaka Power Distribution Company
DR	demand response
DSL	digital subscriber line
DSM	demand side management
DSOPF	dynamic stochastic optimal power flow
DST	decision support tool
DVR	dynamic voltage restorer
EAC	equal area criterion
EMS	energy management system
ES	expert system
EV	electric vehicle
FACTS	flexible AC transmission systems
FAN	field area network
FDI	false data injection
FDLF	fast decoupled load flow
FIPA	Foundation for Intelligent Physical Agents
FLISR	fault location, isolation and service restoration
G2V	grid-to-vehicle
GIS	geographic information system
GPS	Global Positioning System
GS	Gauss-Seidel
HAN	home area network
HES	head-end system
HMI	human-machine interface
HVDC	high-voltage direct current
Hz	hertz (cycles per second)
ICT	information and communication technology
IDAPS	intelligent distributed autonomous power system
IEC	International Electrotechnical Commission
IED	intelligent electronic device
IEEE	Institute of Electrical and Electronics Engineers
IoT	Internet of Things
IPP	independent power producer
IT	information technology
JADE	Java Agent DEvelopment framework

KCL	Kirchhoff's current law
KVL	Kirchhoff's voltage law
kW	kilowatt
kWh	kilowatt-hour
LED	light-emitting diode
LFC	load frequency control
LNG	liquefied natural gas
LTE	Long Term Evolution (4G cellular)
MAS	multi-agent system
MDMS	meter data management system
MGCC	microgrid central controller
MTU	master terminal unit (master station)
MVA	mega volt-ampere
MW	megawatt
NAN	neighbourhood area network
NERC	North American Electric Reliability Corporation
NESCO	Northern Electricity Supply Company
NIST	National Institute of Standards and Technology (USA)
NLDC	National Load Despatch Centre
NR	Newton-Raphson
OLTC	on-load tap changer
OMS	outage management system
OpenADR	Open Automated Demand Response
OPF	optimal power flow
OT	operational technology
PCC	point of common coupling
PDC	phasor data concentrator
pf	power factor
PGCB	Power Grid Company of Bangladesh (now Power Grid Bangladesh PLC)
PLC	programmable logic controller (or power line communication, by context)
PMU	phasor measurement unit
PPS	pulse per second
PQ	power quality (also PQ bus = P and Q specified)
PSS	power system stabiliser
PT	potential (voltage) transformer
pu	per unit (value / base value)
PV	photovoltaic (solar cells); also PV bus = P and $\|V\|$ specified

RF	radio frequency
RI	random index
ROCOF	rate of change of frequency
RTU	remote terminal unit
SCADA	supervisory control and data acquisition
SCOPF	security-constrained optimal power flow
SG	smart grid
SMES	superconducting magnetic energy storage
SMPS	switched-mode power supply
SREDA	Sustainable and Renewable Energy Development Authority
SSA	static security assessment
STATCOM	static synchronous compensator
SVC	static var compensator
TCR	thyristor-controlled reactor
TCSC	thyristor-controlled series capacitor
THD	total harmonic distortion
Tk	Taka
TLS	Transport Layer Security
ToU	time of use
TSC	thyristor-switched capacitor
UPFC	unified power flow controller
UPS	uninterruptible power supply
UTC	Coordinated Universal Time
V2G	vehicle-to-grid
V2H	vehicle-to-home
VEE	validation, editing and estimation
VFD	variable frequency drive
VIPI	voltage instability proximity index
VPN	virtual private network
WAMPAC	wide area monitoring, protection and control
WAMS	wide area monitoring system
WAN	wide area network
WiMAX	Worldwide Interoperability for Microwave Access (IEEE 802.16)
WLAN	wireless local area network (Wi-Fi)
WPAN	wireless personal area network
WZPDCL	West Zone Power Distribution Company Ltd